

Growth-Coupled Carotenoids Production Using Adaptive Laboratory Evolution

Luis H. Reyes and Katy C. Kao

Abstract

Adaptive laboratory evolution is a powerful technique for strain development. However, the target phenotypes using this strategy have been limited by the required coupling of the phenotype-of-interest with fitness or survival, and thus adaptive evolution is generally not used to improve product formation. If the desired product confers a benefit to the host, then adaptive evolution can be an effective approach to improve host productivity. In this book chapter, we describe an effective adaptive laboratory evolution strategy for improving product formation of carotenoids, a class of compounds with antioxidant potential, in the yeast *Saccharomyces cerevisiae*.

Key words Adaptive laboratory evolution, Yeast, Carotenoids, Antioxidants, Strain improvement

1 Introduction

Adaptive laboratory evolution (ALE) is recognized as a powerful method for improvement of several phenotypes of industrial strains without requiring extensive knowledge of underlying molecular basis of these phenotypes [1–3]. The main caveat with using ALE is that it requires the desired trait to be directly linked with a growth benefit to the microorganism. In ALE, adaptive mutants (strains with improved fitness) arise spontaneously and expand in a population in environments with defined selective pressures. The design of suitable selective pressures is the foundation for successful ALEs. There is a saying in the ALE field that, in colloquial terms, “*you get what you select for*”, which captures the importance of choosing the proper selective pressure for directing ALE.

Engineered strains with increased productivity of a desired compound usually exhibit reduced fitness. During whole-cell biocatalysis using engineered strains, the metabolic intermediates necessary for cell growth and maintenance are depleted, being redirected for biosynthesis of desired compounds. This drainage of energy and biomass precursors, including nucleotides and amino

acids, imposes a metabolic load to the cell, generally resulting in a reduction of biomass-related proteins [4, 5]. This negative impact on cell growth in the production of synthetic and secondary metabolites hinders the use of ALE for improvement of their production. The key in designing an ALE experiment for increased synthetic metabolite production lies in how much its production benefits cell growth. However, since the production of desired compounds confers a fitness reduction, it poses an evolutionary disadvantage and a barrier for the use of ALE for strain development. This barrier may be overcome if an appropriate selective pressure can be designed to counteract the metabolic burden in favor of growth advantage, making ALE a potentially powerful tool to develop better producers.

In this chapter we describe a methodology designed for the improvement of carotenoids production in *Saccharomyces cerevisiae*. Carotenoids, also known as tetraterpenoids, are highly hydrophobic organic pigments that are naturally produced by several organisms from plants to bacteria. These compounds are of particular interest due to their antioxidant properties and pigmentation, with uses in nutraceuticals, cosmetics, and the food industry. The biosynthesis of all carotenoids uses isopentenyl diphosphate (IPP) as precursor; the biosynthetic routes are described elsewhere [6]. Extensive metabolic engineering efforts have been applied to develop *S. cerevisiae* strains for the heterologous production of various carotenoids [6–8]. IPP is also a precursor of other isoprenoids, either via MVA (mevalonic acid) pathway in eukaryotes, or the DXP (1-deoxy-D-xylulose 5-phosphate; also known as methylerythritol phosphate) pathway found in several bacteria. In *S. cerevisiae*, the MVA pathway is predominantly used for ergosterol (provitamin D2) biosynthesis, which is an essential component of the yeast membrane and provides membrane permeability and fluidity [9]. Thus, the heterologous production of carotenoids directly affects growth of the fungal host cell due to the redirected IPP flux towards carotenoids production instead of ergosterol biosynthesis. Indeed, prolonged propagation of engineered carotenogenic yeast in normal culture conditions leads to no improvement in carotenoids productivity and in some cases a net loss of productivity (data not shown). To develop an ALE strategy for improving carotenoids production in yeast, we theorized that oxidative stress could be used as a driving force for the directed evolution of *S. cerevisiae* for enhanced carotenoids production. Upon first identifying the exact condition that allows the production of carotenoids to confer protection against oxidative stress (hydrogen peroxide challenge), our prior work successfully improved heterologous carotenoids production in *S. cerevisiae* [1].

While the work described here is specific for carotenoids production in *Saccharomyces cerevisiae*, this approach can potentially be applied to improve production of other compounds with

antioxidant potential. This method has been recently used to increase 3'-hydroxygenistein production in *Pichia pastoris* and to identify overproducers of lycopene in *Escherichia coli* [10, 11].

2 Materials

2.1 Biological and Chemical Materials

1. Target strain previously engineered for the production of the desired compound with antioxidant properties to be used for adaptive laboratory evolution.
2. Growth media for the course of the evolutionary experiment. Defined media (Yeast Nitrogen Base YNB) is preferred in most cases since it allows better control of nutrients and other conditions. However rich media (Yeast-Peptone-Dextrose YPD) may be used as necessary.
3. Inhibitor stocks. The inhibitor described in this chapter is hydrogen peroxide. Hydrogen peroxide (stock concentration 35% w/w) was used in order to apply oxidative stress to the microbial cultures, although the chosen inhibitor is highly dependent of the endogenous resistance of the microorganism and the requirements of the evolution experiment. Inhibitor stocks should be filter-sterilized.
4. 50% (v/v) sterile glycerol for cryogenic storage of evolving populations and isolated mutants.
5. 425–600 μm acid-washed glass beads (Sigma) for cell disruption.
6. Dodecane anhydrous $\geq 99\%$ (Sigma) for the extraction of the hydrophobic β -carotene and other carotenoids.

2.2 Equipment

1. Bench-top centrifuges.
2. Isothermal incubators and shakers.
3. Spectrophotometer.
4. Microplate reader (e.g., TECAN Infinite[®] M200) for high-throughput analysis, with capabilities of survey scans from OD₃₅₀ to OD₅₅₀.
5. Optical microscope (e.g., Zeiss Axio Scope.A1) for visual inspection of morphology. For yeast, a 40 $\times$ phase contrast objective (for a total of 400 $\times$ magnification combined with the 10 $\times$ magnification of the eye piece) is sufficient. For bacteria, a 100 $\times$ phase contract objective is needed.
6. Culture tubes and/or Erlenmeyer flasks.
7. Disruptor Genie[®] Cell Disruptor (Scientific Industries).

Table 1
List of primers used in this study

Name	Sequence	Notes
NEO_f	5' - TTA AAA AAA TCC TTC TCT TGT CTC ATG CCA ATA AGA TCA ATC AGC TCA GCT TCA CAA ATG CGG ATC <u>CCC GGG TTA ATT AA</u> - 3'	The first 60 bp of each primer correspond to homologous regions in the yeast genome, flanking the <i>CTT1</i> gene. The following 20 bp (underlined) are complementary to the NEO cassette.
NEO_r	5' - TAT AAT TAC GAA TAA TTA TGA ATA AAT AGT GCT GCC TTA ATT GGC ACT TGC AAT GGA CCA GAA <u>TTC GAG CTC GTT TAA AC</u> - 3'	
KO.Ver_f	5' - ATT CGA CGT AGC CTG GAC AC - 3'	Primers used for the verification via PCR of the <i>ctt1Δ</i> knockout strains
KO.Ver_r	5' - TAA TCG TTG AGT TCA TGC CG - 3'	

2.3 Molecular Biology

1. Strain genotype: *S. cerevisiae* strain GSY1136 (Mat α , ura3-52, gal + in S288c background, YBR209W::Act1p-GFP-Act1t-URA3) [12].
2. Primers used are listed in Table 1.
3. Plasmids used: YIplac211YB/I/E* shuttle vector [6] for the production of carotenoids in yeast, and pFA6a-kanMX6 (Addgene plasmid # 39296 [13]) for the deletion of the *CTT1* gene.

3 Methods

The most commonly used strategy for adaptive laboratory evolution is continuous exposure to the chosen stressor, either via *serial batch transfers* or *continuous cultures* [14]. However, this may not always be the best strategy, especially when the production of the desired product poses a metabolic burden (*see Note 1*). In the following section we introduce a modified strategy for adaptive laboratory evolution experiments using a *Periodic Challenge/Recovery Scheme*. This modified method can be implemented when the growth benefit of the produced antioxidant compound is not sufficiently large to compensate for the metabolic load imposed on the microorganism; such is the case for carotenoids production in yeast. Below, we describe in detail the necessary steps for using this methodology in order to increase the carotenoids production in *S. cerevisiae*. Finally, some notes concerning contamination and instances where the described methodology may not be suitable are including in Subheading 4.

3.1 Construction of Carotenoids-Producing Strains

The methods described below correspond to the modification of the *S. cerevisiae* strain GSY1136, however they can be modified for the production of carotenoids in other yeast strains.

1. Since the GSY1136 strain used in this work contains the URA3 marker, it was necessary to excise it in order to create a uracil auxotrophic strain. We recommend using the anti-metabolite 5-fluoroorotic acid (5-FOA) as counter-selection to select for the loss of the URA3 marker, as described elsewhere [15]. This procedure was used to generate the uracil auxotrophic strain GSY1136 *ura3Δ*.
2. Digest the plasmid YIplac211YB/I/E* using the restriction endonuclease *StuI* and purify the linearized plasmid.
3. Transform 1 μg of linearized plasmid into GSY1136 *ura3Δ* using the lithium-acetate procedure or via electroporation to generate the carotenogenic yeast strain GSY1136 YIplac211YB/I/E*.
4. Plate the transformed cells onto solid YNB without amino acids (or just without uracil supplementation) and incubate at 30 °C for 3 days.
5. The correct constructs can be visually verified since the produced carotenoids have a characteristic red color.

3.2 Construction of Catalase-Deficient Strains

The rationale behind using ALE to increase carotenoids production is based on the antioxidant properties of these compounds. *S. cerevisiae* possesses two catalases (CTT1 and CAT1) [16]. It has been demonstrated that the cytosolic catalase CTT1 is essential for protecting the yeast cell against hydrogen peroxide stress [17]. Therefore, it is necessary to delete CTT1 to significantly reduce the native H₂O₂ detoxification mechanisms in yeast, in order to increase the antioxidant benefits of carotenoids. The outline for the deletion of CTT1 via homologous recombination in the yeast strain GSY1136 YIplac211YB/I/E*, is described below; however, alternative methods of disrupting the catalase gene in other yeast strains can also be used. For a different microbial platform, the primers should be modified accordingly.

1. Amplify the NEO cassette from the plasmid pFA6a-kanMX6 using the primers *NEO_f* and *NEO_r*.
2. Purify the amplicon and transform it into competent yeast cells using the lithium-acetate procedure or via electroporation.
3. Plate the cells onto solid media supplemented with geneticin (G418) at 200 μg/ml for selection.
4. Incubate the cells for 2 days or until colonies are observed.
5. Verify knockout strains (GSY1136 YIplac211YB/I/E* *ctt1Δ*) by PCR, using the primers *KO.Ver_f* and *KO.Ver_r*.

6. In our studies, we did not observed significant differences in growth rates between the parental and *ctt1Δ* strains in YNB + glucose or YPD, incubated at 30 °C.

3.3 Periodic Challenge/Recovery Scheme

Serial batch transfers have been routinely used to carry out adaptive evolution experiments [2, 18]. This method involves the serial transfer of evolving populations into media containing an inhibitor used as selective pressure, and the concentration of this inhibitor is generally increased periodically (ramped-up) (*see Note 2*). Initially we hypothesized that by using serial batch transfers, as depicted in Fig. 1a, to perform the evolution experiments, the antioxidant action of carotenoids will alleviate the oxidative stress caused by mild H₂O₂ concentrations close to the MIC₅₀ (defined as the minimal inhibitory concentration [MIC] of an inhibitor necessary to reduce growth of a target microorganism by 50%). However, we discovered that when cells were actively growing, the accumulation of carotenoids within the cell was not sufficient for alleviating H₂O₂ stress; and instead of the intended directed evolution towards increased carotenoids production, the productivity of the population actually decreased over time (data not shown). It is likely that in this case, since the production of the desired metabolite imposes a metabolic burden to the cell, the continuous exposure of a low level of oxidative stress was not sufficient for the benefit of carotenoids production to overcome the metabolic burden.

Preliminary experiments were carried out to determine the exact condition in which the production of carotenoids confers a benefit to the producing host. Since carotenoids are constitutively produced using the pTDH3 promoter [6], the maximum intracellular yield of the bio-product was expected to occur at the end of the exponential growth phase. We compared the cell viability of cell cultures from early-, mid-, and late-exponential phase exposed to high (lethal) concentrations of hydrogen peroxide in short periods of time, and a higher survival of yeast cells was observed in cells challenged during late-exponential growth.

Therefore, a new strategy using *Periodic Challenge/Recovery* was implemented. This modified methodology aims to fully exploit the antioxidant capabilities of carotenoids. Instead of continuous exposure of the cells in a stressful environment, as in usual evolutionary experiments, the populations were cultured in the absence of stressor until late-exponential/early stationary phase for two reasons: first to generate a larger amount of biomass and second to maximize the accumulation of carotenoids inside the cells. Periodic challenge of carotenoids-producing cells in the appropriate growth phase with high concentrations of hydrogen peroxide for short periods of time (shocking periods) (Fig. 1b) is then used to select for mutants with enhanced carotenoids production, as better producers are expected to exhibit increased survival. During the periodic challenge, ~95% of the population die, but cells producing

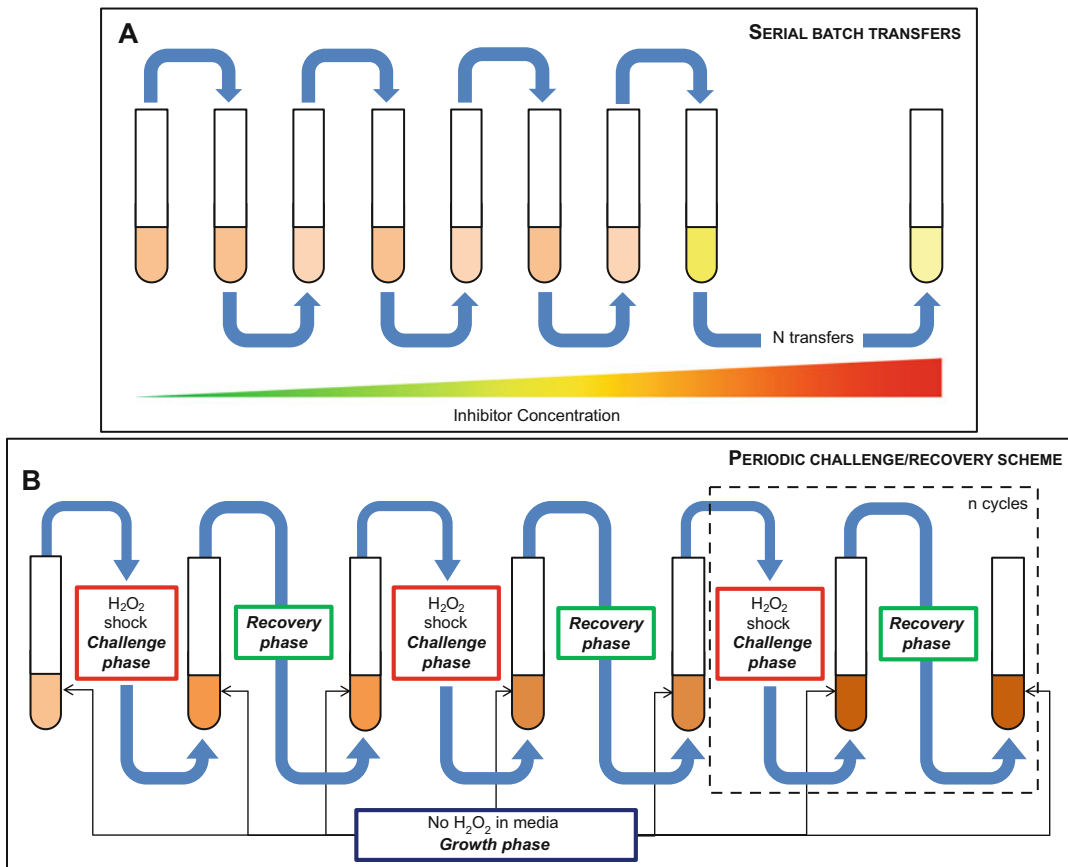


Fig. 1 (a) The *Serial Batch Transfer Method*, the most commonly used strategy for adaptive laboratory evolution where the microorganisms are continuously exposed to a chosen stressor. The selective pressure may be increased throughout the course of the adaptive laboratory evolution. (b) The modified method called *Periodic Challenge/Recovery Scheme*. This two-phase strategy uses a *challenge* phase using high concentrations of hydrogen peroxide to select for cells with larger antioxidants properties, and a *recovery* phase without the stressor to allow the cells more time to recover after peroxide shock

the highest amounts of carotenoids have better chance of survival. This *challenge* phase is followed by a *recovery* phase to allow the cells more time to recover from the hydrogen peroxide shock. This approach should significantly reduce the contribution of metabolic burden from the equation. Preliminary experiments should be carried out for the specific host producer strain, and the specific stressor used to determine the appropriate strategy to use, especially when to impose the challenge phase.

3.4 Determination of Sensitivity of Production Host to Oxidative Stress

Before starting any adaptive laboratory evolution experiment, it is indispensable to determine the inhibitory effect of the compound used to create the selective pressure. For this reason, it is crucial to determine the amount of H₂O₂ for the ALE. This amount is based on determining the effect of the selective pressure on cell viability; a

bottleneck of 5% survival is an arbitrary, but reasonable starting point. It is recommended to use a non-producing strain as control to identify if the effect of β -carotene on H_2O_2 alleviation is large enough to confer a fitness advantage to the producer strain.

1. Streak out the strain of interest onto appropriated agar plates. Incubate the plates at permissible temperature until colonies are visible.
2. Pick one colony for every set of biological replicates and inoculate 3 ml of liquid media (same as one to use in evolution experiments). Grow cells at the appropriated conditions to maximize carotenoids production.
3. Normalize cultures by cell density and transfer 500 μl of the normalized culture to microcentrifuge tubes.
4. Add various amounts of hydrogen peroxide to the samples, to get concentrations ranging from 0.5 M to 3.0 M. Mix samples by vortex for 3 s. The amount of hydrogen peroxide needed may vary depending of the strain being studied.
5. Allow the reaction to proceed for 30 min. It is recommended to keep the tubes lids open in a sterile environment (e.g., laminar flow hood) due to the continuous release of O_2 from the reaction.
6. Pellet cells in a bench-top centrifuge at $3000 \times g$ for 5 min.
7. Remove supernatant and resuspend cells in fresh liquid media.
8. Plate cells onto agar plates at several dilutions and determine cell viability. A concentration of hydrogen peroxide ensuring a cell survival around a target frequency (e.g., 5%) may be used as the starting point for the implementation of periodic challenge/recovery strategy.

3.5 Periodic Challenge/Recovery Experiments

1. Streak out strain of interest onto agar plates. Grow cells at permissible temperature until colonies are visible.
2. Pick one colony for every set of technical replicates and inoculate it into liquid media. Replicated cultures from independent colonies are necessary in case of events of genetic drift due to the large bottleneck that the periodic challenge/recovery experiments represent.
3. Make glycerol stocks of the overnight cultures to save the parental strain as control for subsequent analyses. This is especially important when identification of mutations associated with improved productivity is desired, since jackpot mutations may already be present in the inoculum.
4. Prepare X test tubes or Erlenmeyer flasks with γ ml of media.
 - (a) The number of replicates (X) depends on the number of biological and technical replicates (ideally a minimum of

three independent colonies as biological replicates with two technical replicates each).

- (b) The media volume (γ) depends on culture vessel used in the experiment. As a rule of thumb, the volume of the culture should be 20% of the total volume of the culture vessel to ensure adequate mixing and aeration.
5. Inoculate the cultures with the overnight culture using 1–2% (v/v) inoculum (e.g., for a 100 ml culture, use 1–2 ml of overnight culture).
6. Incubate the cultures at the appropriated conditions to maximize carotenoids production.
7. Once the samples reach the desired condition (late-exponential phase in our experience):
 - (a) Quantify the carotenoids concentration using the protocol described in Subheading 3.5.1.
 - (b) Prepare X sterile 1.5 ml microcentrifuge tubes for hydrogen peroxide shock.
 - (c) Prepare glycerol stocks of the evolving populations using 67% culture and 33% by volume 50% glycerol. Store at 80 °C for later analysis.
 - (d) Prepare X more tubes (or Erlenmeyer flasks) with γ ml of liquid media.
8. Add the amount of hydrogen peroxide determined from Subheading 3.3 to the samples. Mix samples by vortex for 3 s.
9. Allow the reaction to proceed for 30 min taking into account the recommendations from Subheading 3.3.
10. Pellet cells in a bench-top centrifuge at $3000 \times g$ for 5 min.
11. Remove supernatant and resuspend cells in fresh liquid media.
12. Inoculate X new cultures with 5% (v/v) inoculum from treated or untreated (control or during recovery phase) cells and cultivate until the cultures reach stationary phase.
13. Repeat **steps 6, 7 and 12** to allow the cells to recover after the hydrogen peroxide shock treatment (serial passage without a hydrogen peroxide challenge).
14. Repeat **steps 6–12** for a new round of peroxide challenge.
15. Repeat **steps 13 and 14** until the desired experimental phenotypic goals have been reached or for a set number of generations (*see* **Notes 3 and 4**).

3.5.1 Quantification of Carotenoids

1. Transfer 250 μ l of culture to a 2 ml microcentrifuge tube with screw cap.
2. Collect cells by centrifugation at $11,000 \times g$ for 2 min. Remove the supernatant avoiding pellet disturbance.

3. Add 250 μl of 425–600 μm acid-washed glass beads (Sigma) and 1 ml of dodecane.
4. Lyse cells using an analog Disruptor Genie[®] Cell Disruptor (Scientific Industries) or equivalent. By trial and error, we determined that by treating the samples twice for 6 min, we ensure maximum cell disruption and carotenoids recovery. However, the disruption time is strain dependent. We encourage the researcher to determine the appropriated disruption time for the strain-of-interest.
5. Separate cell debris and glass beads from the supernatant by centrifugation at $18,000 \times g$ for 2 min. If pelleted cells remain pigmented, this indicates that extraction was not complete; therefore, the disruption time should be increased.
6. Transfer 200 μl of the supernatant to a Corning[®] 96-well clear-bottom plate for quantification.
7. Ideally, HPLC analysis should be used to obtain exact quantifications of carotenoids (*see Note 5*). However, in the absence of available tools, a spectrophotometer or microplate reader (TECAN Infinites[®] or similar) can be used to quantify total carotenoids using a survey scan from OD₃₅₀ to OD₅₅₀, in order to determine any shifts in the spectrum. The relative total carotenoids production is determined by calculating the area under the curve of the survey scan, using the parental strain as reference. The maximum absorbance of β -carotene occurs at 454 nm, thus quantification of β -carotene is determined by the absorption at 454 nm. It is vital to have a standard curve for β -carotene quantification. This can be generated using commercially available β -carotene (Enzo Life Sciences) at OD₄₅₄.

4 Notes

1. *It is crucial to determine the level of toxicity of the bio-product of interest.* In this chapter we described how to compensate for the metabolic load imposed by the biosynthesis of carotenoids; carotenoids are generally not toxic to the cells. In this case, we were able to design an effective selective pressure to direct evolution towards the desired phenotype. If the compound-of-interest not only imposes a metabolic burden, but is also toxic to the microorganism, then the described methodology may not be adequate to overcome those issues. In those cases, it is recommended to first increase host tolerance to the toxic bio-product, and once a tolerant strain is obtained, the described methodology can be implemented to increase its production.
2. Chemical mutagenesis can be used to accelerate the evolutionary process by increasing the mutation rate of the system.

However, it is recommended to use it cautiously to avoid getting too many mutations per strain to prevent beneficial mutants with improved productivity to acquire deleterious mutations.

3. It is imperative to periodically track evolving populations for microbial and fungal contamination. Commonly used techniques for this purpose includes: PCR genotyping, Chromocult[®] Coliform Agar, Gram staining, and visual inspection of culture under microscope.
4. Maintenance of sterile conditions is mandatory to prevent contamination. Every accessory in direct contact with the evolving populations must be sterilized.
5. The protocol for carotenoids quantification we described may not be accurate if different carotenoids with overlapping absorbance spectra are produced; although it is a simple method for qualitative assessment of strain performance. For more accurate quantification of carotenoids, HPLC [19, 20] can be used.

References

1. Reyes LH, Gomez JM, Kao KC (2014) Improving carotenoids production in yeast via adaptive laboratory evolution. *Metab Eng* 21:26–33. doi:[10.1016/j.ymben.2013.11.002](https://doi.org/10.1016/j.ymben.2013.11.002)
2. Almario MP, Reyes LH, Kao KC (2013) Evolutionary engineering of *Saccharomyces cerevisiae* for enhanced tolerance to hydrolysates of lignocellulosic biomass. *Biotechnol Bioeng* 110(10):2616–2623. doi:[10.1002/bit.24938](https://doi.org/10.1002/bit.24938)
3. Reyes LH, Almario MP, Winkler J, Orozco MM, Kao KC (2012) Visualizing evolution in real time to determine the molecular mechanisms of n-butanol tolerance in *Escherichia coli*. *Metab Eng* 14(5):579–590. doi:[10.1016/j.ymben.2012.05.002](https://doi.org/10.1016/j.ymben.2012.05.002)
4. Bentley WE, Mirjalili N, Andersen DC, Davis RH, Kompala DS (1990) Plasmid-encoded protein: the principal factor in the “metabolic burden” associated with recombinant bacteria. *Biotechnol Bioeng* 35(7):668–681. doi:[10.1002/bit.260350704](https://doi.org/10.1002/bit.260350704)
5. Rozkov A, Avignone-Rossa CA, Ertl PF, Jones P, O’Kennedy RD, Smith JJ, Dale JW, Bushell ME (2004) Characterization of the metabolic burden on *Escherichia coli* DH1 cells imposed by the presence of a plasmid containing a gene therapy sequence. *Biotechnol Bioeng* 88(7):909–915. doi:[10.1002/bit.20327](https://doi.org/10.1002/bit.20327)
6. Verwaal R, Wang J, Meijnen JP, Visser H, Sandmann G, van den Berg JA, van Ooyen AJ (2007) High-level production of beta-carotene in *Saccharomyces cerevisiae* by successive transformation with carotenogenic genes from *Xanthophyllomyces dendrorhous*. *Appl Environ Microbiol* 73(13):4342–4350. doi:[10.1128/AEM.02759-06](https://doi.org/10.1128/AEM.02759-06)
7. Li Q, Sun Z, Li J, Zhang Y (2013) Enhancing beta-carotene production in *Saccharomyces cerevisiae* by metabolic engineering. *FEMS Microbiol Lett* 345(2):94–101. doi:[10.1111/1574-6968.12187](https://doi.org/10.1111/1574-6968.12187)
8. Xie W, Lv X, Ye L, Zhou P, Yu H (2015) Construction of lycopene-overproducing *Saccharomyces cerevisiae* by combining directed evolution and metabolic engineering. *Metab Eng* 30:69–78. doi:[10.1016/j.ymben.2015.04.009](https://doi.org/10.1016/j.ymben.2015.04.009)
9. Nielsen J (2009) Systems biology of lipid metabolism: from yeast to human. *FEBS Lett* 583(24):3905–3913. doi:[10.1016/j.febslet.2009.10.054](https://doi.org/10.1016/j.febslet.2009.10.054)
10. Xu X, Jin W, Jiang L, Xu Q, Li S, Zhang Z, Huang H (2016) A high-throughput screening method for identifying lycopene-overproducing *E. coli* strain based on an antioxidant capacity assay. *Biochem Eng J* 112:277–284. doi:[10.1016/j.bej.2016.04.032](https://doi.org/10.1016/j.bej.2016.04.032)
11. Wang TY, Tsai YH, Yu IZ, Chang TS (2016) Improving 3'-hydroxygibberic acid production in recombinant *Pichia pastoris* using periodic hydrogen peroxide-shocking strategy. *J Microbiol Biotechnol* 26(3):498–502. doi:[10.4014/jmb.1509.09013](https://doi.org/10.4014/jmb.1509.09013)

12. Kao KC, Sherlock G (2008) Molecular characterization of clonal interference during adaptive evolution in asexual populations of *Saccharomyces cerevisiae*. *Nat Genet* 40 (12):1499–1504. doi:[10.1038/ng.280](https://doi.org/10.1038/ng.280)
13. Bahler J, Wu JQ, Longtine MS, Shah NG, McKenzie A 3rd, Steever AB, Wach A, Philippsen P, Pringle JR (1998) Heterologous modules for efficient and versatile PCR-based gene targeting in *Schizosaccharomyces pombe*. *Yeast* 14(10):943–951. doi:[10.1002/\(SICI\)1097-0061\(199807\)14:10<943::AID-YEA292>3.0.CO;2-Y](https://doi.org/10.1002/(SICI)1097-0061(199807)14:10<943::AID-YEA292>3.0.CO;2-Y)
14. Winkler J, Reyes LH, Kao KC (2013) Adaptive laboratory evolution for strain engineering. *Methods Mol Biol* 985:211–222. doi:[10.1007/978-1-62703-299-5_11](https://doi.org/10.1007/978-1-62703-299-5_11)
15. Boeke JD, LaCroute F, Fink GR (1984) A positive selection for mutants lacking orotidine-5'-phosphate decarboxylase activity in yeast: 5-fluoro-orotic acid resistance. *Mol Gen Genet* 197(2):345–346
16. Switala J, Loewen PC (2002) Diversity of properties among catalases. *Arch Biochem Biophys* 401(2):145–154. doi:[10.1016/S0003-9861\(02\)00049-8](https://doi.org/10.1016/S0003-9861(02)00049-8)
17. Martins D, English AM (2014) Catalase activity is stimulated by H₂O₂ in rich culture medium and is required for H₂O₂ resistance and adaptation in yeast. *Redox Biol* 2:308–313. doi:[10.1016/j.redox.2013.12.019](https://doi.org/10.1016/j.redox.2013.12.019)
18. Winkler JD, Garcia C, Olson M, Callaway E, Kao KC (2014) Evolved osmotolerant *Escherichia coli* mutants frequently exhibit defective N-acetylglucosamine catabolism and point mutations in cell shape-regulating protein MreB. *Appl Environ Microbiol* 80(12):3729–3740. doi:[10.1128/AEM.00499-14](https://doi.org/10.1128/AEM.00499-14)
19. Breithaupt DE (2004) Simultaneous HPLC determination of carotenoids used as food coloring additives: applicability of accelerated solvent extraction. *Food Chem* 86(3):449–456. doi:[10.1016/j.foodchem.2003.10.027](https://doi.org/10.1016/j.foodchem.2003.10.027)
20. Kaiser P, Surmann P, Vallentin G, Fuhrmann H (2007) A small-scale method for quantitation of carotenoids in bacteria and yeasts. *J Microbiol Methods* 70(1):142–149. doi:[10.1016/j.mimet.2007.04.004](https://doi.org/10.1016/j.mimet.2007.04.004)